47 Gravity waves

The various particles of matter which make up the universe act on each other in various ways. If they are charged, for instance, they might attract or repel each other with a force known as the electromagnetic force. If they have mass, they attract each other by means of the gravitational force (there does not seem to be a repulsive version). The particles in an atomic nucleus are affected by two types of force, descriptively called the strong nuclear force and the weak nuclear force! These four forces can act over a distance (though the two nuclear ones are *very* short range). The electromagnetic force has a particle associated with it, called a photon, and the repulsion of two electrons can be reinterpreted as the exchange of this particle between them, as shown in figures 1a and 1b. In their attempts to find a unified field theory which links the four forces in a single structure, physicists have suggested, with some success, that there should be a particle associated with each force; for the strong force it is a 'gluon', for the weak force a 'W boson' (there is also a Z boson) and for the gravitational force a 'graviton'. To call photons, and the rest, particles is simplistic – they also show wave properties. The search for gluons and bosons has been carried out using particle accelerators at very high energies, but the search for gravitons from distant astronomical events such as supernovae or colliding neutron stars is being pursued in a very different way. Figure 2 shows the idea of a proposed graviton detector.

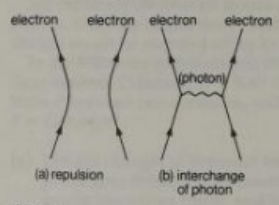

Figure 1

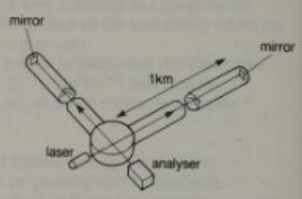

Figure 2

The two arms are evacuated metal cylinders 1 km long, with a laser beam travelling along each one and back from a mirror at the far end. The laser beams are then combined with some of the original laser light to produce interference patterns. A graviton should cause a tiny change in length of either arm, of only a fraction of the wavelength of light, and this will produce a significant change in the interference pattern. The movements expected are so small that traffic vibration, seismic disturbances, and even the thermal vibration of the atoms of the tubes have to be minimised so that they do not mask the effect. The lasers proposed must also have a constant frequency, with a variation of only 1 mHz in 5×10^{14} Hz. *Two* arms at right angles mean that any time difference between changes in the interference patterns enables the direction of the gravitons to be worked out.

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Data: $\frac{1}{4\pi\varepsilon_o} = 9 \times 10^9\ \mathrm{Nm^2C^{-2}}$

$G = 6.7 \times 10^{-11}\ \mathrm{Nm^2kg^{-2}}$
mass of electron $= 9.1 \times 10^{-31}$ kg
mass of proton $= 1.7 \times 10^{-27}$ kg
charge on electron $= 1.6 \times 10^{-19}$ C
charge on a proton $= +1.6 \times 10^{-19}$ C
speed of light $= 3 \times 10^8\ \mathrm{ms^{-1}}$

Relationship:
force between charges q_1 and q_2 distance r apart $= \frac{1}{4\pi\varepsilon_o}\frac{q_1q_2}{r^2}$

force between masses m_1 and m_2 distance r apart $= \frac{Gm_1m_2}{r^2}$

(a) Calculate the electrical force between an electron and a proton if they are 10^{-10} m apart. (3)
(b) Calculate the gravitational force between the electron and proton in part (a). (3)
(c) What is the ratio between the electrical and the gravitational forces in (a) and (b)? (2)
(d) The nucleus of a gold atom is made up of 79 protons and 118 neutrons, and alpha particles fired at a gold leaf can be strongly repelled from a nucleus (as in Geiger and Marsden's famous experiment). How do these facts tell you that the strong nuclear force is stronger than the electromagnetic force and that it is short range? (6)
(e) Give reasons why a remote part of Scotland has been suggested for the graviton detector, and why the arms should be cooled to a very low temperature. (6)
(f) Why does the frequency of the laser light proposed to measure the length of the arms have to be so stable? (2)
(g) Calculate the wavelength of the light. (2)
(h) If a shift of 1/20 of a wavelength can be detected in the interference pattern, what percentage change in length of an arm can be measured? (3)
(i) From which directions could gravitons be arriving if the changes were detected simultaneously? (3)

Total 30 marks

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CIE 9702 Physics (Paper 4 style): Gravitational-wave detector

Passage

Physicists model four fundamental interactions: gravitational, electromagnetic, strong and weak. The electromagnetic force between two point charges is given by

$$F_e = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2},$$

while the gravitational force between two point masses is

$$F_g = G\frac{m_1m_2}{r^2}.$$

In a simplified gravitational-wave detector, two evacuated metal arms are each 1.0 km long. A laser beam of frequency $f = 5.0 \times 10^{14}$ Hz is split so that light travels along both arms and reflects from mirrors at the ends. The returning beams interfere. A passing graviton/gravitational-wave signal causes a tiny difference in arm lengths, changing the interference pattern.

To detect this effect, the arms must remain extremely stable. It is suggested that random thermal motion and external vibrations can hide the signal unless the apparatus is cooled and isolated. A detectable shift corresponds to a path-length change of about half a wavelength.

Use the following data:

$$\frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9\,\mathrm{N\ m^2C^{-2}}, \quad G = 6.67 \times 10^{-11}\,\mathrm{N\ m^2kg^{-2}},$$

$$m_e = 9.11 \times 10^{-31}\,\mathrm{kg}, \quad m_p = 1.67 \times 10^{-27}\,\mathrm{kg}, \quad e = 1.60 \times 10^{-19}\,\mathrm{C}, \quad c = 3.0 \times 10^8\,\mathrm{m\ s^{-1}}.$$

Questions

1. Show that the magnitude of the electric force between an electron and a proton separated by $r = 1.0 \times 10^{-10}\,\mathrm{m}$ is approximately $2.3 \times 10^{-8}\,\mathrm{N}$. [3]

2. Calculate the gravitational force between the same electron and proton at the same separation. [3]

3. Determine the ratio F_e/F_g and comment on what this implies for attempts to detect individual gravitons in laboratory conditions. [3]

4. For the laser in the detector:

(a) calculate the wavelength, [2]

(b) estimate the minimum detectable arm-length change if the detector can resolve a shift of $\lambda/2$. [2]

5. Explain why cooling the detector arms and mirrors is important, and why high-frequency laser light helps in resolving tiny displacement changes. [5]

6. Open-ended. A new research group has funding to improve this detector by changing *one* main design parameter (for example: arm length, laser frequency, vibration isolation, cooling system, or mirror technology). Propose one improvement and justify, with physics reasoning, how it increases sensitivity to gravitational waves. Include at least two limitations or trade-offs. [7]

Total: 25 marks

Suggested answers (concise marking guide)

1.

$$F_e = 9.0 \times 10^9 \times \frac{(1.60 \times 10^{-19})^2}{(1.0 \times 10^{-10})^2} = 2.3 \times 10^{-8}\,\mathrm{N}.$$

2.

$$F_g = 6.67 \times 10^{-11} \times \frac{(9.11 \times 10^{-31})(1.67 \times 10^{-27})}{(1.0 \times 10^{-10})^2} \approx 1.0 \times 10^{-47}\,\mathrm{N}.$$

3.

$$\frac{F_e}{F_g} \approx \frac{2.3 \times 10^{-8}}{1.0 \times 10^{-47}} \approx 2 \times 10^{39}.$$

Electromagnetic effects dominate strongly; gravitational effects at particle scale are extremely weak and hard to isolate.

4(a).

$$\lambda = \frac{c}{f} = \frac{3.0 \times 10^{8}}{5.0 \times 10^{14}} = 6.0 \times 10^{-7}\,\mathrm{m}.$$

4(b).

$$\Delta L_{\min} \approx \frac{\lambda}{2} = 3.0 \times 10^{-7}\,\mathrm{m}.$$

5. Cooling reduces random thermal vibrations (noise); isolation reduces seismic/traffic vibrations; both improve signal-to-noise ratio. Higher frequency light gives shorter wavelength, so a smaller displacement causes a measurable phase change.

6. Example valid response: increase arm length. For a given strain h, displacement signal $\Delta L = hL$, so larger L gives larger absolute displacement. Trade-offs: cost, land/infrastructure, alignment complexity, added environmental noise coupling. Other valid choices earn credit if justified physically with limitations.

Variant 2 (Higher difficulty): Advanced gravitational-wave analysis

Passage

A second detector design is proposed to improve sensitivity to very small strains. The detector has perpendicular arms of length $L = 3.0\,\text{km}$ and uses laser light of frequency $f = 7.5 \times 10^{14}\,\text{Hz}$. A gravitational wave passing perpendicular to the detector plane causes opposite fractional changes in arm lengths, producing a phase shift in the recombined light.

For small strain h, the differential arm change may be taken as

$$\Delta L \approx hL.$$

If the optical system can detect a phase shift corresponding to an effective path change of $\delta = \lambda/200$, then the smallest measurable strain can be estimated from

$$h_{\text{min}} \approx \frac{\delta}{L}.$$

A compact binary source at distance R emits nearly isotropically with gravitational-wave power P. A simplified intensity model gives

$$I = \frac{P}{4\pi R^2}.$$

For a wave speed close to c, the energy density is approximately

$$u \approx \frac{I}{c}.$$

Use $c = 3.0 \times 10^8\,\text{m s}^{-1}$ and $1\,\text{ly} = 9.46 \times 10^{15}\,\text{m}$.

Questions (Variant 2)

1. Calculate the laser wavelength for $f = 7.5 \times 10^{14}\,\text{Hz}$. [2]

2. Determine the minimum detectable path change δ if the phase sensitivity is $\lambda/200$. [2]

3. Hence estimate the minimum detectable strain h_{min} for $L = 3.0\,\text{km}$. [3]

4. A source emits power $P = 1.2 \times 10^{30}\,\text{W}$ and is at distance $R = 2.0 \times 10^8\,\text{ly}$.
Calculate:

(a) the intensity I at Earth, [3]

(b) the corresponding energy density u. [2]

5. The detector is upgraded from 3.0 km arms to 4.0 km arms, while all other parameters remain unchanged. Determine the factor by which h_{min} changes. [2]

6. Explain two reasons why practical detector sensitivity may be much worse than the simple estimate from Question 3, even if optical precision is high. [4]

7. Open-ended. Two competing upgrade proposals are considered:

- Proposal A: increase laser frequency by 40%.
- Proposal B: reduce effective noise floor by a factor of 2 using better suspension and cryogenic cooling.

Recommend which proposal is likely to yield the larger real-world gain in detection capability. Support your argument using both quantitative estimates and qualitative physics reasoning. [7]

Total (Variant 2): 25 marks

Suggested answers (Variant 2)

1.

$$\lambda = \frac{c}{f} = \frac{3.0 \times 10^8}{7.5 \times 10^{14}} = 4.0 \times 10^{-7}\,\mathrm{m}.$$

2.

$$\delta = \frac{\lambda}{200} = \frac{4.0 \times 10^{-7}}{200} = 2.0 \times 10^{-9}\,\mathrm{m}.$$

3.

$$h_{\min} \approx \frac{\delta}{L} = \frac{2.0 \times 10^{-9}}{3.0 \times 10^3} \approx 6.7 \times 10^{-13}.$$

4(a).

$$R = 2.0 \times 10^8 \times 9.46 \times 10^{15} = 1.892 \times 10^{24}\,\mathrm{m},$$

$$I = \frac{1.2 \times 10^{30}}{4\pi(1.892 \times 10^{24})^2} \approx 2.7 \times 10^{-20}\,\mathrm{W\ m^{-2}}.$$

4(b).

$$u \approx \frac{I}{c} = \frac{2.7 \times 10^{-20}}{3.0 \times 10^8} \approx 9.0 \times 10^{-29}\,\mathrm{J\ m^{-3}}.$$

5. Since $h_{\min} \propto 1/L$,

$$\frac{h'_{\min}}{h_{\min}} = \frac{3.0}{4.0} = 0.75.$$

So the minimum detectable strain decreases to 75% of its previous value (a 25% improvement).

6. Any two valid points, e.g. thermal noise, seismic/vibration noise, quantum shot noise, radiation-pressure noise, mirror coating losses, residual gas effects, imperfect alignment/calibration.

7. Indicative judgement: Proposal B is often stronger in practice because broadband noise reduction improves signal-to-noise across the detection band, whereas increasing frequency by 40% only reduces λ (and thus displacement scale) by about 1/1.4. A well-argued quantitative comparison and limitations discussion earns full credit.